

Requirements Specification

Project Name: KanBan

Team: Uffe, Bhuwan, Michael

Introduction

This project aims to develop a Kanban software application, a scheduling system for managing tasks. A user interface displays cards displaying tasks that are grouped according to their progress in the development lifecycle.

This full-stack project aims to deliver a front-end and a back-end. Additionally, we aim to keep data persistent via a database, and possibly online access via cloud service providers.

Technical Information

Category	Details
Frontend	Angular, TypeScript, Node.js, Tailwind
Backend	Python, FastAPI, SQLAlchemy
Database	SQLite (or PostgreSQL), Alembic for migrations
Cloud	Azure / AWS

Functional Requirements

- Add a task
- Update a task
 - Editing the text
 - Moving the task

Other Features

- Drag and drop on the user interface
- Remove a task
- Add name to task
- Adjustable number of columns/headers
- Databases
- Cloud (AWS/Azure)
- User accounts and roles
- User permissions

Out of Scope

- User authentication
- Asynchronous/multi-user boards

- Adding a new board
- Swim lanes (horizontal dividers)

Division of Roles

Team Member	Responsibilities
Uffe	Project management, Backend development (Python), Unit testing
Bhuwan	Frontend development (Angular), Backend integration
Michael	Data types & databases (SQL), Cloud migration (AWS/Azure), Software architecture

Plan

Day	Task
Mandag	Planning: user stories, tech stack requirement specification
Tirsdag	CV and job applications
Onsdag	Define API endpoints, MVP development (Add task, Edit text, Moving task)
Torsdag	MVP estimated finished
Fredag	Unit testing, documentation, Main product development (drag and drop, remove task, add name to task)
Mandag	Unit testing, documentation, Main product development, Nice to have
Tirsdag	Unit testing, documentation, Main product development, Nice to have
Onsdag	CV and job applications
Torsdag	Nice to have, project wrap up
Fredag	Project wrap up, Presentation

Mockup

In this mock-up, users can add tasks by clicking the add button within the relevant columns. We placed the “Add New Task” button in the area where users expect the new tasks to be.

In general, users should be able to modify elements by clicking on them. For example, tasks can be moved around with drag-and-drop, and can be deleted by dropping the task into the bin. Simulating a physical board like this is intended

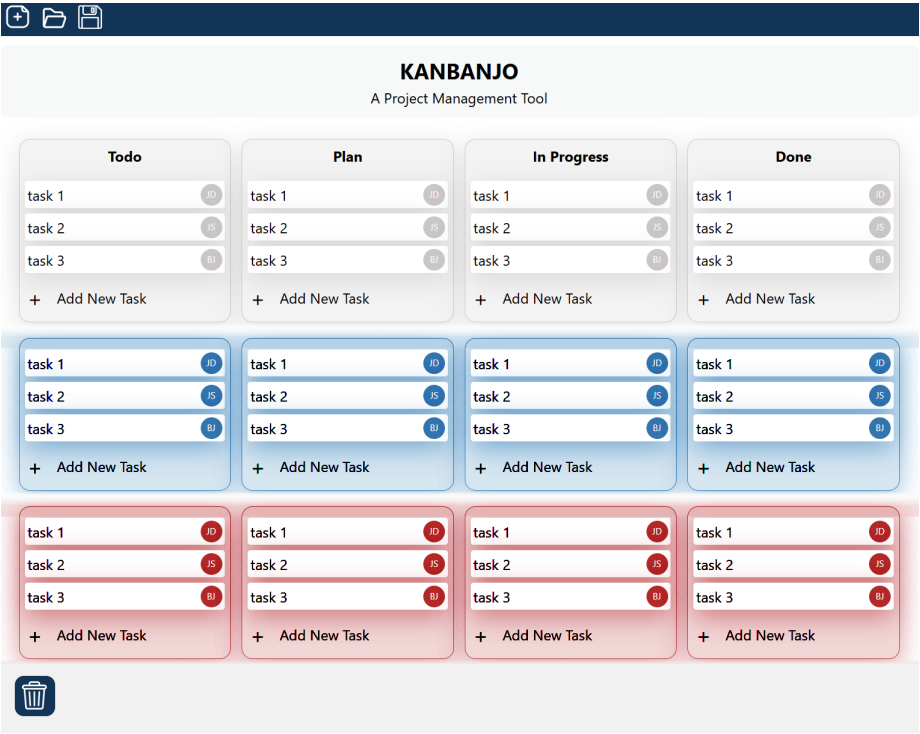


Figure 1: Mockup

to make the interaction more intuitive, and it is an expected behaviour of kanban boards.

Text (such as columns headers or task content) can be edited by clicking on the text, and then typing in new contents. By making elements editable by clicks and drags, we can minimize clutter such as buttons and menus.

Creating and loading are handled in the top left corner. We placed these items in the top left area because users expect this as a convention.

Finally, the color coding of rows represent the swim lanes. The swim lanes can be used to define categories horizontally – for example, the graphics team can have tasks grouped separately from the audio team.